

$$\frac{p_{(k)}}{p} \leq \frac{k}{m_0 + j - 1} \frac{m_0 + j - 1}{(m + 1)p} q^*. \quad (7)$$

When conditioning on $P'_{(m_0)} = p$, the P'_i/p for $i = 1, 2, \dots, m_0 - 1$ are distributed as $m_0 - 1$ independent $U(0, 1)$ random variables, and the p_i/p for $i = 1, 2, \dots, j$ are numbers corresponding to false null hypotheses between 0 and 1. Using inequality (7) to test the $m_0 + j - 1 = m' \leq m$ hypotheses is equivalent to using procedure (1), with the constant $\{(m_0 + j - 1)/(m + 1)p\}q^*$ taking the role of q^* . Applying now the induction hypothesis, we have

$$E(Q | P'_{(m_0)} = p, P_{m_0+1} = p_1, \dots, P_m = p_{m_1}) \leq \frac{m_0 - 1}{m_0 + j - 1} \frac{m_0 + j - 1}{(m + 1)p} q^* = \frac{m_0 - 1}{(m + 1)p} q^* \quad (8).$$

The bound in inequality (8) depends on p , but not on the segment $p_j < p < p_{j+1}$ for which it was evaluated, so

$$\begin{aligned} \int_{p^*}^1 E(Q | P'_{(m_0)} = p, P_{m_0+1} = p_1, \dots, P_m = p_{m_1}) f_{P'_{(m_0)}}(p) dp &\leq \int_{p^*}^1 \frac{m_0 - 1}{(m + 1)p} q^* m_0 p^{(m_0-1)} dp \\ &= \frac{m_0}{m + 1} q^* \int_{p^*}^1 (m_0 - 1) p^{(m_0-2)} dp = \frac{m_0}{m + 1} q^* \{1 - p^{(m_0-1)}\}. \end{aligned} \quad (9)$$

Adding inequalities (6) and (9) completes the proof of the lemma.

REFERENCES

- Godfrey, K. (1985) Comparing the means of several groups. *New Engl. J. Med.*, **311**, 1450-1456.
- Hochberg, Y. (1988) A sharper Bonferroni procedure for multiple tests of significance. *Biometrika*, **75**, 800-803.
- Hochberg, Y. and Benjamini, Y. (1990) More powerful procedures for multiple significance testing. *Statist. Med.*, **9**, 811-818.
- Hochberg, Y. and Tamhane, A. (1987) *Multiple Comparison Procedures*. New York: Wiley.
- Holm, S. (1979) A simple sequentially rejective multiple test procedure. *Scand. J. Statist.*, **6**, 65-70.
- Hommel, G. (1988) A stagewise rejective multiple test procedure based on a modified Bonferroni test. *Biometrika*, **75**, 383-386.
- Neuhaus, K. L., Von Essen, R., Tebbe, U., Vogt, A., Roth, M., Riess, M., Niederer, W., Forycki, F., Wirtzfeld, A., Maeurer, W., Limbourg, P., Merx, W. and Haerten, K. (1992) Improved thrombolysis in acute myocardial infarction front-loaded administration of Alteplase: results of the rt-PA-APSAC patency study (TAPS). *J. Am. Coll. Card.*, **19**, 885-891.
- Pocock, S. J., Hughes, M. D. and Lee, R. J. (1987) Statistical problems in reporting of clinical trials. *J. Am. Statist. Ass.*, **84**, 381-392.
- Rom, D. M. (1990) A sequentially rejective test procedure based on a modified Bonferroni inequality. *Biometrika*, **77**, 663-665.
- Saville, D. J. (1990) Multiple comparison procedures: the practical solution. *Am. Statistn*, **44**, 174-180.
- Simes, R. J. (1986) An improved Bonferroni procedure for multiple tests of significance. *Biometrika*, **73**, 751-754.
- Smith, D. E., Clemens, J., Crede, W., Harvey, M. and Gracely, E. J. (1987) Impact of multiple comparisons in randomized clinical trials. *Am. J. Med.*, **83**, 545-550.
- Sorici, B. (1989) Statistical "discoveries" and effect size estimation. *J. Am. Statist. Ass.*, **84**, 608-610.
- Spjøtvoll, E. (1972) On the optimality of some multiple comparison procedure. *Ann. Math. Statist.*, **43**, 398-411.